



Thermopower and Nernst effect in graphene in a magnetic field

Joseph G. Checkelsky and N. P. Ong

Department of Physics, Princeton University, Princeton, New Jersey 08544, USA

(Received 13 January 2009; revised manuscript received 2 August 2009; published 27 August 2009)

We report measurements of the thermopower S and Nernst signal S_{yx} in graphene in a magnetic field H . Both quantities show strong quantum oscillations vs the gate voltage V_g . Our measurements for Landau levels of index $n \neq 0$ are in quantitative agreement with the edge-current model of Girvin and Jonson (GJ). The inferred off-diagonal thermoelectric conductivity α_{xy} is consistent in magnitude with GJ's prediction. At the Dirac point ($n=0$), however, the width of the peak in α_{xy} is very narrow. We discuss the physical meaning of the peak width and the puzzle raised by α_{xy} for states at the Dirac point.

DOI: [10.1103/PhysRevB.80.081413](https://doi.org/10.1103/PhysRevB.80.081413)

PACS number(s): 73.63.Bd, 72.15.Jf, 73.43.Fj

In graphene, the electronic states at the corners K and K' of the first Brillouin zone have a linear dispersion described by the two-dimensional (2D) massless Dirac Hamiltonian. In a magnetic field $\mathbf{H}$, graphene exhibits the integer quantum Hall effect (QHE).^{1–4} As the chemical potential μ is raised, successive filling of Landau levels (LLs) leads to a Hall conductivity that is quantized as $\sigma_{xy} = (4e^2/h)[n + \frac{1}{2}]$, where e is the electron charge, h is Planck's constant, and n is the Landau-level index. The QHE in graphene exhibits novel features not encountered in quadratic systems based on GaAs, especially in the $n=0$ LL. To date, these issues have been explored nearly exclusively by resistance measurements.^{1–7} By contrast, the thermoelectric response is much less investigated. An applied temperature gradient $-\nabla T \parallel \hat{\mathbf{x}}$ (with $\mathbf{H} \parallel \hat{\mathbf{z}}$) generates an electric field $\mathbf{E}$ in the sample with components $E_x = S_{xx}(-\partial_x T)$ and $E_y = S_{yx}(-\partial_x T)$, where S_{ij} is the thermopower tensor. A thermopower measurement yields E_x (hence S_{xx}), while the Nernst effect yields E_y and S_{yx} . By measuring the elements of S_{ij} and σ_{ij} (the conductivity tensor), we may determine the charge current density $\mathbf{J}$ produced by $-\nabla T$. Its transverse component J_y is especially interesting in the QHE regime. According to Girvin and Jonson (GJ),^{8,9} the ratio $J_y/|\nabla T|$ has the quantized value $(k_B e/h) \ln 2$ when $n \neq 0$ (k_B is Boltzmann's constant). However, nothing is known about how J_y behaves in the $n=0$ LL (where the GJ theory fails). We investigate this little-explored quantization in graphene. [For later reference, J_y is given by $J_y = \alpha_{yx}(-\nabla T)$, where the thermoelectric conductivity tensor α_{ij} relates $-\nabla T$ to $\mathbf{J}$.]

A most unusual feature of the thermoelectric response of a QHE system is that the thermopower displays a large peak at each LL ($n \neq 0$), whereas the Nernst signal is small. In the geometry treated by GJ (Refs. 8 and 9) [Fig. 1(a)], the 2D sample is of finite width along $\hat{\mathbf{x}}$ but is infinite along $\hat{\mathbf{y}}$. As we approach either edge, the energy $E_n(x)$ of each LL rises very steeply (bold curve). At $T=0$, edge currents I_y exist at the intersections of E_n with the chemical potential μ (open circles). In a gradient $-\nabla T$, the magnitude of I_y is larger at the warmer edge compared with the cooler edge because of increased occupation of states above μ . As we show below, the current difference δI_y is quantized. For a finite sample, δI_y produces a quantized Hall potential $V_H = (h/e^2) \delta I_y$ whose gradient lies $\parallel (-\nabla T)$. Consequently, when μ is aligned with the LL energy in the bulk E_n , $S_{xx} \sim E_x$ displays a large peak, whereas the Nernst signal $S_{yx} \sim E_y$ is small.

As mentioned, the calculation of GJ does not extend to the $n=0$ LL. Moreover, the nature of the edge currents at $n=0$ (Dirac point in graphene) is the subject of current debate.^{6,7,10–13} What are the profiles of S and S_{yx} ? We have measured S_{ij} and R_{ij} to a maximum H of 14 T at 20 and 50 K. Our results reveal that, at 9 T, the thermoelectric response in graphene already falls in the quantum regime at 50 K. The inferred off-diagonal term α_{xy} is a series of peaks independent of n , B , and T . The peak values are compared with $(k_B e/h) \ln 2$ below.

To measure the thermopower, we used a technique developed by Kim's group for carbon nanotubes.¹⁴ The graphene sample was exfoliated onto a thin layer (300 nm) of SiO₂ grown on n -doped Si wafer. A strip of gold evaporated near one end of the sample served as the heat source [labeled "Heater" in Fig. 1(d)]. To measure the gradient $-\nabla T$, we evaporated the two gold strips labeled "Therm1" and "Therm2." Their four-probe resistances were measured to four significant figures between 10 and 300 K. Thereafter, they could be employed as resistance thermometers for sensing the local temperature at the two ends (with resolution $\sim \pm 1$ mK above 10 K). They were also used as current leads for measurements of R_{ij} . The voltage signals V_{ij} were detected by the four gold lines labeled as "Signal Leads." For a temperature difference $\delta T \sim 10$ mK, the uncertainty in the gradient $-\nabla T$ was $\pm 10\%$, mostly due to the uncertainty in the voltage lead spacing ($\sim 2 \mu\text{m}$). To improve the signal-to-noise ratio, we used an alternating heater current of frequency $\omega/2\pi \sim 3$ Hz to produce an alternating gradient. The thermoelectric response was detected at 2ω , with a phase shift of -90° .

In our geometry with $-\nabla T \parallel \hat{\mathbf{x}}$, $\mathbf{H} \parallel \hat{\mathbf{z}}$, the charge current density $\mathbf{J}$ is $J_i = \sum_j [\sigma_{ij} E_j + \alpha_{ij} (-\partial_j T)]$ ($i=x, y$). Setting $\mathbf{J}=0$, we solve for $\mathbf{E}$ and obtain

$$E_i = - \sum_{k,j} \rho_{ik} \alpha_{kj} (-\partial_j T) = \sum_j S_{ij} (-\partial_j T), \quad (1)$$

with $\rho_{ij} = R_{ij}$ the 2D resistivity tensor. The thermopower $S = -E_x/|\nabla T|$ equals $\rho_{xx} \alpha_{xx} + \rho_{yx} \alpha_{xy}$ ($S > 0$ for hole doping), while the Nernst signal is given by $S_{yx} = \rho_{xx} \alpha_{xy} - \rho_{yx} \alpha_{xx}$ (with $\rho_{xx} = \rho_{yy}$).

Inverting Eq. (1), we may calculate the tensor α_{ij} from measured quantities. We have



磁场中石墨烯的热电效应和能斯特效应

Joseph G. Checkelsky和N. P. Ong

美国新泽西州普林斯顿市普林斯顿大学物理系, 邮编08544

(2009年1月13日收到; 修订稿于2009年8月2日收到; 2009年8月27日发表)

我们报告了石墨烯在磁场 H 中热电势 S 和能斯特信号 S_{yx} 的测量结果。这两个量均显示出与栅极电压 V_g 相关的强烈量子振荡。我们对指数 $n=0$ 的朗道能级的测量结果与Girvin和Jonson (GJ) 的边缘电流模型定量一致。推断出的非对角热导率 α_{xy} 的量级与GJ的预测相符。然而在狄拉克点 ($n=0$) 处, α_{xy} 的峰宽非常窄。我们讨论了该峰宽的物理意义以及 α_{xy} 对狄拉克点态提出的谜题。

DOI: [10.1103/PhysRevB.80.081413](https://doi.org/10.1103/PhysRevB.80.081413)

PACS编号: 73.63. Bd, 72.15. Jf, 73.43. Fj

在石墨烯中, 第一布里渊区角 K 和 K' 处的电子态具有由二维 (2D) 无质量狄拉克哈密顿量描述的线性色散。在磁场 H 作用下, 石墨烯表现出整数量子霍尔效应 (QHE)。随着化学势 μ 的升高, 朗道能级 (LLs) 的连续填充导致霍尔电导量子化为 $\sigma_{xy} = (4e^2/h) [n + \frac{1}{2}]$, 其中 e 是电子电荷, h 是普朗克常数, n 是朗道能级指数。石墨烯中的 QHE 表现出基于砷化镓的二次系统所未遇到的新型特征, 特别是在 $n=0$ LL中。迄今为止, 这些问题几乎完全通过电阻测量来探索。相比之下, 热电响应研究较少。施加的温度梯度 $-\nabla T \parallel \hat{x}$ (与 $H \parallel \hat{z}$) 在样品中产生电场 E , 其分量为 $E_x = S_{xx} (-\partial_x T)$ 和 $E_y = S_{yx} (-\partial_x T)$, 其中 S_{ij} 为热电势张量。热电势测量得到 E_x (即 S_{xx}), 而能斯特效应则产生 E_y 和 S_{yx} 。通过测量 S_{ij} 和 σ_{ij} (电导率张量) 的元素, 我们可以确定由 $-\nabla T$ 产生的电荷电流密度 J 。其横向分量 J_y 在 QHE 区域尤为有趣。根据Girvin 和 Jonson (GJ)^{8,9}, 当 $n=0$ 时 (k_B 为玻尔兹曼常数), 比值 $J_y/|\nabla T|$ 具有量子化值 $(k_B e/h) \ln 2$ 。然而, 目前尚不清楚 J_y 在 $n=0$ LL (即GJ理论失效的情形) 中的行为。我们对此进行了初步研究。研究了石墨烯中的量子化现象。其 [供后续参考, J_y 是热电表达式为 $J_y = \alpha_{yx} (-\nabla T)$, 其中梯导率-度张量 α_{ij} 将 $-\nabla T$ 与 J 相关联。

QHE 系统热电响应的一个非常特殊特征是, 热电势在每个LL ($n=0$) 处均呈现显著峰值, 而能斯特信号则较弱。在GJ研究的几何结构中 (参考文献8和9) [图1(a)], 二维样品沿 $\hat{x}$ 但沿 $\hat{y}$ 方向是无限的。当我们接近任一边缘时, 每个LL的能量 $E_n(x)$ 急剧上升 (粗曲线)。在 $T=0$ 时, 边缘电流 I_y 存在于 E_n 与化学势 μ 的交点处 (空圆)。在梯度 $-\nabla T$ 下, 由于 μ 以上能态的占据增加, I_y 的幅值在较热边缘比较冷边缘更大。如下所示, 电流差 δI_y 是量子化的。对于有限样品, δI_y 产生量子化霍尔势 $V_H = (h/e2) \delta I_y$, 其梯度为 $\Pi (-\nabla T)$ 。因此, 当 μ 与体相 E_n , $S_{xx} \sim E_x$ 中的LL能量对齐时, $S_{xx} \sim E_x$ 显示出一个大峰, 而能斯特信号 $S \sim E$ 较小。

如前所述, GJ的计算并不适用于 $n=0$ 的LL。此外, $n=0$ 处 (石墨烯中的狄拉克点) 边缘电流的本质仍是当前争论的焦点。^{6,7,10-13} S 和 S_{yx} 的分布特征是什么? 我们已测量了 S_{ij} 和 R_{ij} , 在20和50处的最大 H 为14 T。

K. 我们的研究表明, 在9 T磁场下, 石墨烯的热电响应在50 K时已进入量子化区域。推断出的非对角项 α_{xy} 呈现一系列与 n 、 B 和 T 无关的峰。峰值与下文 $(k_B e/h) \ln 2$ 进行对比。

为测量热电势, 我们采用了Kim团队为碳纳米管开发的技术。¹⁴将石墨烯样品剥离到生长在 n 型掺杂硅片上的 SiO_2 薄层 (300 nm) 上。在样品一端附近蒸发的金条作为热源 [图1(d)中标注为 “Heater”]。为测量梯度 $-\nabla T$, 我们蒸发了标记为 “Therm1” 和 “Therm2” 的两条金条。它们的四探针电阻在10至300 K之间被测量到四位有效数字。此后, 它们可作为电阻温度计用于检测两端的局部温度 (在10 K以上分辨率约为 ± 1 mK)。它们还被用作测量 R_{ij} 的电流引线。电压信号 V_{ij} 由标记为 “Signal Leads” 的四条金线检测。对于温度差 δT 约10 mK, 梯度 $-\nabla T$ 的不确定度为 $\pm 10\%$, 主要源于电压引线间距的不确定度 (约 μm)。为提高信噪比, 我们采用频率 $\omega/2\pi \sim 3$ Hz的交变加热电流产生交变梯度。热电响应在 2ω 处检测, 相位偏移为 -90° 。

在我们的几何结构中, $-\nabla T \parallel \hat{x}$, $H \parallel \hat{z}$, 充电电流密度 J 为 $J_i = \sum_j [\sigma_{ij} E_j + \alpha_{ij} (-\partial_j T)]$ $[i=x, y]$ 。设 $J=0$, 求解 E 并得到

$$E_i = - \sum_{k,j} \rho_{ik} \alpha_{kj} (-\partial_j T) = \sum_j S_{ij} (-\partial_j T), \quad (1)$$

其中 $\rho_{ij} = R_{ij}$ 为二维电阻率张量。热电势 $S = -E_x/|\nabla T|$ 等于 $\rho_{xx} \alpha_{xx} + \rho_{yx} \alpha_{xy}$ (空穴掺杂时 $S > 0$), 而能斯特信号由 $S_{yx} = \rho_{xx} \alpha_{xy} - \rho_{yx} \alpha_{xx}$ 给出 (其中 $\rho_{xx} = \rho_{yy}$)。

通过反转等式 (1), 我们可以根据测量量计算张量 α_{ij} 。我们有

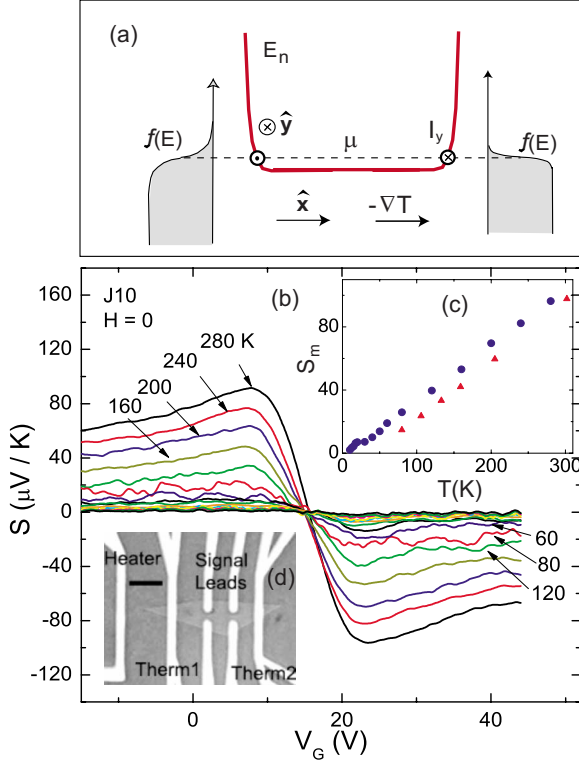


FIG. 1. (Color online) (a) The effect of $-\nabla T$ on the edge currents I_y in a QHE system ($n \neq 0$). The energy E_n of a LL (bold curve) increases very steeply at the sample edges. The dashed line is the chemical potential μ . If $H_z > 0$, I_y is negative (positive) at the left (right) edge as indicated by open circles. The magnitude $|I_y|$ is larger at the warmer edge. Fermi-Dirac distributions $f(E)$ are sketched at the sides. (b) Curves of thermopower $S = -S_{xx}$ vs gate voltage V_g in sample J10 at selected T . The curves are antisymmetric about the Dirac point which occurs at the offset voltage $V_0 = 15.5$ V. The peak value S_m is nominally linear in T from 25 to 300 K (c). Less complete data from sample K59 are also plotted. A photo of sample J10 (faint polygon) is shown in (d). A microheater as well as thermometers (therm) and signal leads are patterned with electron-beam lithography. The black scale bar is $3 \mu\text{m}$.

$$\alpha_{xx} = -(\sigma_{xx}E_x + \sigma_{xy}E_y)/|\nabla T|,$$

$$\alpha_{xy} = (-\sigma_{xy}E_x + \sigma_{xx}E_y)/|\nabla T|. \quad (2)$$

Under field reversal ($\mathbf{H} \rightarrow -\mathbf{H}$), S is symmetric, whereas S_{yx} is antisymmetric. For each curve taken in field, we repeat the measurement with $\mathbf{H}$ reversed. All curves of S and S_{yx} reported here have been (anti)symmetrized with respect to $\mathbf{H}$. As for charge-inversion symmetry, we expect the sign of S to be odd with respect to the shifted gate voltage $V'_g \equiv V_g - V_0$, with V_0 the offset voltage ($V'_g = 0$ defines the charge-neutral Dirac point). However, the Nernst signal S_{yx} is symmetric in V'_g .

Figure 1(b) shows traces of S vs V_g at selected T with $H = 0$. The thermopower S changes sign as V_g crosses the Dirac point, becoming positive (negative) on the hole (electron) side. The peak value S_m is nominally T linear from ~ 20 K to 300 K [Fig. 1(c)].

In sharp contrast to the smooth variation in Fig. 1, the

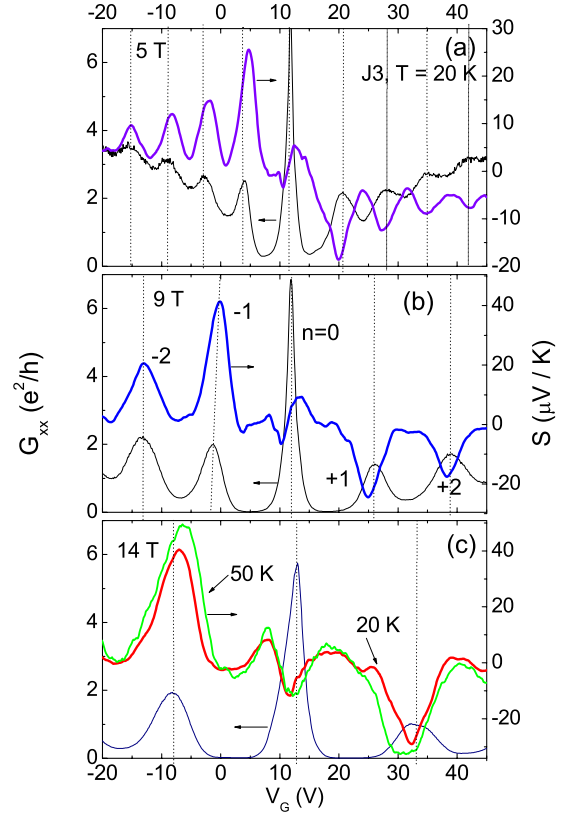


FIG. 2. (Color online) Variation in thermopower S vs V_g (bold curves) and conductance G_{xx} vs V_g (thin curves) in sample J3 at $H = 5, 9$, and 14 T [panels (a), (b), and (c), respectively]. The offset $V_0 = 12.5$ V. All curves were measured at 20 K except for the curve at 50 K in panel (c). Vertical lines locate the maxima of G_{xx} . The LL index n is indicated in panel (b).

curves of S vs V_g at finite H show pronounced oscillations, reflecting Landau quantization of the Dirac states. Figures 2(a)–2(c) display S vs V_g (bold curves) with H fixed at the values $5, 9$ and 14 T, respectively. For comparison, we have also plotted (as thin curves) the corresponding conductance $G_{xx} = \sigma_{xx}$. [With the exception of the curve at 50 K in panel (c), all curves were measured at 20 K.] Whereas at large $|n|$, the peaks in S are aligned with those in G_{xx} [vertical lines in Fig. 2(a)], at $n = \pm 1$, they disagree. In panel a, the peaks for $V'_g < 0$ (hole doping) decrease systematically in magnitude as n increases from 1 to 4 .

In Ref. 8, GJ show that, at each $n \neq 0$, the edge current difference δI_y leads to a quantized α_{yy} , viz.

$$\alpha_{xy}(\mu) = \frac{k_B e}{h} \beta \int_{E_n}^{\infty} d\epsilon (\epsilon - \mu) \left(-\frac{\partial f}{\partial \epsilon} \right), \quad (3)$$

where $\beta = (k_B T)^{-1}$ and $f(\epsilon)$ is the Fermi-Dirac distribution. As μ is varied, α_{xy} rises to a narrow peak with peak value $\alpha_{xy}^{\max} = (k_B e/h) \ln 2 \approx 2.32$ nA/K, attained when μ equals E_n in the bulk. The width of the peak—discussed below—broadens linearly with T .

When combined with σ_{ij} , Eq. (3) implies that at LL n , the thermopower has a peak value

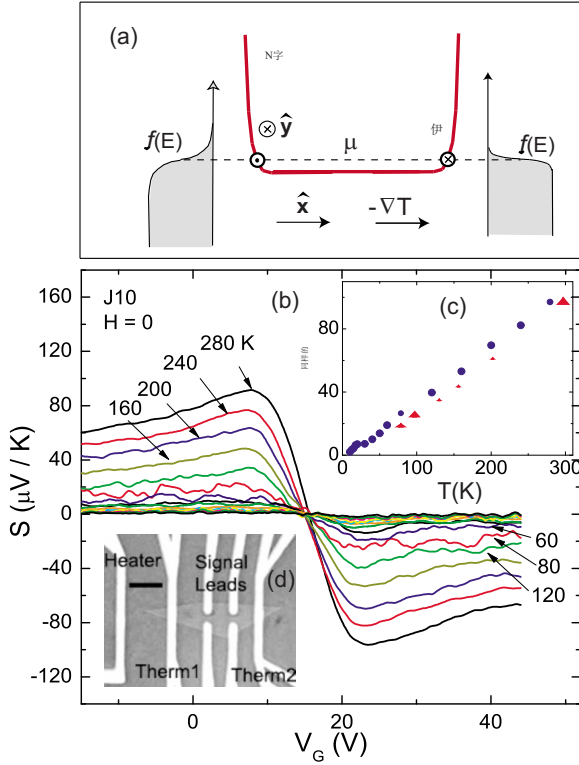


图1. (彩色在线) (a) QHE 系统中 $-\nabla T$ 对边缘电流 I_y 的影响 ($n=0$)。朗道能级 (LL) 的能量 E_n (粗线) 在样品边缘急剧上升。虚线表示化学势 μ 。若 $H>0$, 则 I_y 在左侧 (右侧) 边缘为负 (正), 如空心圆所示。较热边缘的 $|I_y|$ 幅值更大。费米-狄拉克分布 $f(E)$ 在两侧示意。(b) 样品 J10 在选定 T 下热电势 $S = -S_{xx}$ 随栅极电压 V 的变化曲线。曲线呈反对称关于在偏置电压 $V_0 = 15.5$ V 处出现的狄拉克点的描述。峰值 S_m 在 25 至 300 K 范围内 (c) 名义上 T 呈线性关系。K59 样品的不完整数据也已绘制。样品 J10 (淡色多边形) 的照片显示在 (d) 中。微加热器以及温度计 (therm) 和信号引线均采用电子束光刻技术图案化。黑色标尺为 $3 \mu\text{m}$ 。

$$\alpha_{xx} = -(\sigma_{xx}E_x + \sigma_{xy}E_y)/|\nabla T|,$$

$$\alpha_{xy} = (-\sigma_{xy}E_x + \sigma_{xx}E_y)/|\nabla T|. \quad (2)$$

在场反转 ($\mathbf{H} \rightarrow -\mathbf{H}$) 条件下, S 是对称的, 而 S_{yx} 是反对称的。对于场中选取的每条曲线, 我们重复测量 $\mathbf{H}$ 反转后的结果。所有 S 和 S 的曲线在此报告的对称性已相对于 $\mathbf{H}$ 进行了 (反) 对称化处理。关于电荷反转对称性, 我们预期 S 的符号关于偏置电压 Dirac 点 V_0 移位栅极电压 $V'_g = V_g - V_0$ ($V_g = 0$ 定义的特性表现异常。然而, 了电荷中性的能斯特信号 S 是对称的 V' 。

图 1(b) 显示了在选定 T 下 $H=0$ 时 S 随 V_g 的变化轨迹。当 V 穿过狄拉克点, 在空穴 (电子) 侧变为正 (负)。 S_m 名义上从 ~ 20 K 到 300 K 呈 T 线性 [图 1(c)]。

与图 1 中平滑的变化形成鲜明对比的是,

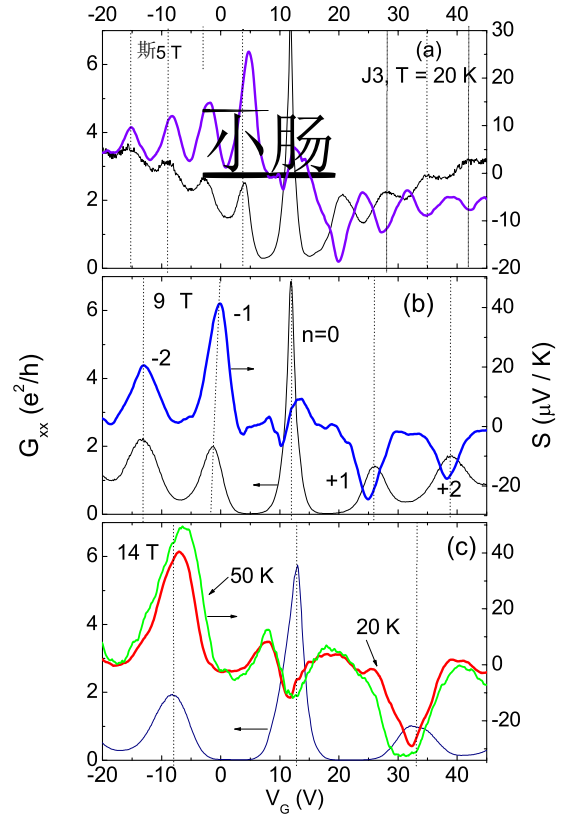


图2. (彩色在线) 样品 J3 在 $H=5, 9$ 和 14 T 时热电势 S 随 V_g 的变化 (粗曲线) 与电导 G_{xx} 随 V_g 的变化 (细曲线) [分别见图 (a)、(b) 和 (c)]。偏移量 $V_0 = 12.5$ V。除图 (c) 中 50 K 曲线外, 所有曲线均在 20 K 下测量。垂直线标示 G_{xx} 的最大值。LL 指数 n 标注于图 (b)。

S 随 V_g 变化的曲线在有限 H 下显示出显著振荡, 反映了狄拉克态的朗道量子化。图 2(a)–2(c) 分别展示了 S 随 V_g 变化 (粗线曲线), 其中 H 固定为 $5, 9$ 和 14 T。为便于比较, 我们还绘制了相应的电导 $G_{xx} = \sigma_{xx}$ (细线曲线)。[除图 (c) 中 50 K 的曲线外, 所有曲线均在 20 K 下测量。] 当 $|n|$ 较大时, S 的峰值与 G_{xx} [图 2(a) 中的垂直线] 对应; 而在 $n = \pm 1$ 时, 它们不一致。在图 a 中, $V_g < 0$ (空穴掺杂) 的峰值随 n 从 1 增加到 4 而系统地减小。

在参考文献 8 中, GJ 表明, 在每个 $n=0$ 时, 边缘电流差 δI_y 导致量子化的 α_{yy} , 即

$$\alpha_{xy}(\mu) = \frac{k_B e}{h} \beta \int_{E_n}^{\infty} d\epsilon (\epsilon - \mu) \left(-\frac{\partial f}{\partial \epsilon} \right), \quad (3)$$

其中 $\beta = (k_B T)^{-1}$, 且 $f(\epsilon)$ 为费米-狄拉克分布。 μ 呈变化趋势, α_{xy} 升至狭窄峰值, 峰值为

$\alpha_{xy}^{\max} = (k_B e/h) \ln 2 \approx 2.32$ nA/K, 当体材料中 μ 等于 E_n 时达到。峰宽 (下文讨论) 随 T 线性展宽。

当与 σ_{ij} 结合时, 等式 (3) 表明在 LL n 处, 热电势具有峰值。

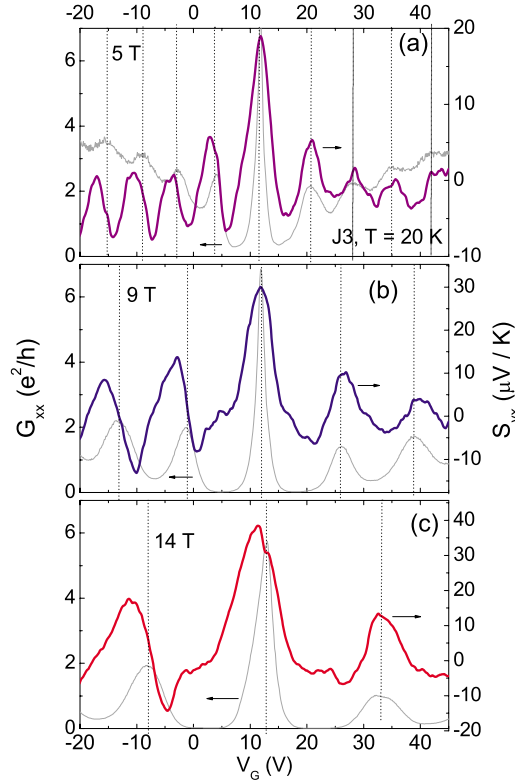


FIG. 3. (Color online) Variation in Nernst signal S_{yx} vs V_g (bold curves) and conductance G_{xx} vs V_g (thin curves) in J3 at the three field values $H=5, 9$, and 14 T [panels (a), (b), and (c), respectively]. All curves were measured at 20 K. Vertical lines locate the maxima of G_{xx} . At the Dirac point ($V'_g=0$), the sign of S_{yx} (i.e., E_y) is positive, the same as for the vortex-Nernst signal in a superconductor (Refs. 15 and 16).

$$S_{peak}(n) = \frac{k_B}{e} \frac{\ln 2}{\left(n + \frac{1}{2}\right)}. \quad (4)$$

At low T , S is independent of H and T .

In Fig. 2, the peak value of S at the $n=-1$ LL increases from 25 at 5 T to $41 \mu V/K$ at both 9 and 14 T. Moreover, as T increases from 20 to 50 K (panel c), the peak increases only weakly ($41-48 \mu V$) in sharp contrast with the T -linear behavior at $H=0$ (Fig. 1, inset). This is consistent with the prediction of GJ that S (at the peak) saturates to a value independent of T and H at sufficiently low T . This saturation contrasts with the T -linear behavior of S in $H=0$ [Fig. 1(c)].

In graphene, however, the Berry phase effects lead to a $\frac{1}{2}$ -integer shift in the LL index.³ In evaluating $\sigma_{xy} \sim \sum_n f(E_n)$, the $\frac{1}{2}$ -integer shift implies that $S_{peak}(n)$ decreases as $k_B \ln 2/(en)$ instead of Eq. (4). The measured values $S_{peak}=41 \mu V/K$ at $n=-1$ at 9 T already exceeds slightly the predicted value $39.7 \mu V/K$ in Eq. (4). Improved measurements on cleaner samples may yield values closer to the predicted value $59.6 \mu V/K$ for Dirac systems. In principle, Eq. (4) provides a way to measure δT on micron-scales with very high resolution—comparable to measurements of resistance.

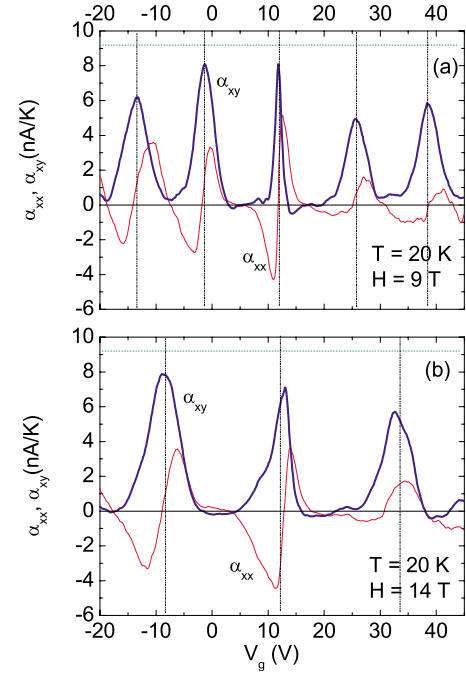


FIG. 4. (Color online) The thermoelectric response functions α_{xx} (faint curve) and α_{xy} (bold) vs V_g at 9 T (a) and 14 T (b) at $T=20$ K calculated from S and S_{yx} [Eq. (2)]. In (a), the width of the peak at $n=0$ is narrower than the others by a factor of ~ 4 . In both panels, vertical lines locate the peaks of G_{xx} . The horizontal dashed line is $(4k_B e/h) \ln 2$.

The curves of the Nernst signal S_{yx} are displayed in Fig. 3 for $H=5, 9$, and 14 T. In the three panels, the dominant feature is the large central peak at the $n=0$ LL which towers over the peaks at levels with $n \neq 0$. (By contrast S in Fig. 2 is suppressed at $n=0$ but large for $n \neq 0$.) We note that the observed large Nernst peak and the strong suppression of S at $n=0$ are strongly inconsistent with the flat-top profiles calculated for a system with quadratic dispersion.⁹ The positive sign of S_{yx} at $n=0$ implies that the Nernst E -field $\mathbf{E}_N$ is parallel to $\mathbf{H} \times (-\nabla T)$ (same sign as in the vortex-Nernst effect in superconductors).^{15,16}

With S and S_{yx} measured, we may use Eq. (2) to calculate the thermoelectric conductivity tensor elements α_{ij} . We display α_{xy} (bold curves) and α_{xx} (thin curves) vs V_g in Fig. 4(a) (for $H=9$ T) and in Fig. 4(b) (14 T). Unlike S and S_{yx} , the peaks in α_{xy} are much narrower and clearly separated by broad intervals in which α_{xy} is nominally zero.

A striking feature of α_{xy} is that its peaks are nominally independent of n . The prediction of the model of GJ (Refs. 8 and 9) is that [in the geometry of Fig. 1(a)], the transverse current J_y displays a series of narrow pulses whenever μ aligns with E_n in the bulk (for $n \neq 0$). The peak current per unit temperature gradient depends only on the combination of universal constants $k_B e/h$. Figure 4 confirms this behavior for $H=9$ and 14 T. The pulses have an average value ~ 75 nA/K which is within 20% of $(gk_B e/h) \ln 2$ with $g=4$ (horizontal dashed line). Given the large uncertainty in estimating the gradient $-\nabla T$ between the voltage leads, the agreement is satisfactory.

However, the peak of α_{xy} at $n=0$ raises several interesting

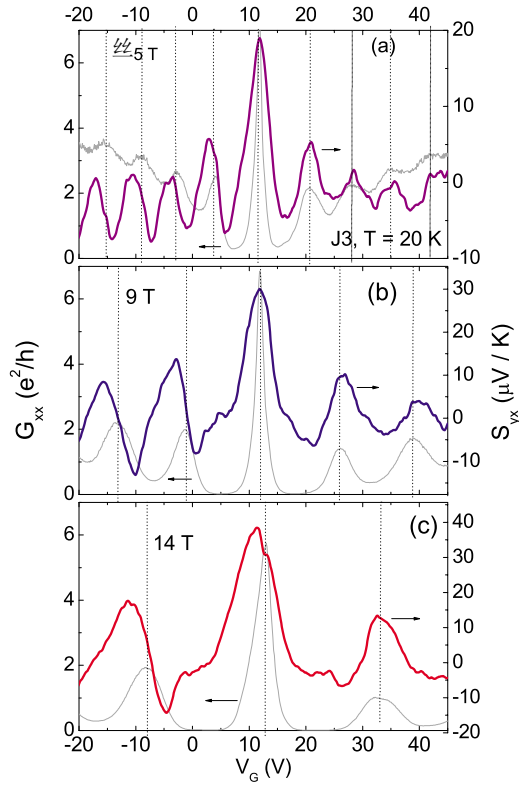


图3. (彩色在线) J3在三个磁场值 $H=5$ 、 9 和 14 T时的能斯特信号 S_{yx} 随 V_g 的变化(粗曲线)及电导 G_{xx} 随 V_g 的变化(细曲线)[分别见图(a)、(b)和(c)]。所有曲线均在 20 K下测量。垂直线标示 G_{xx} 的最大值。在狄拉克点($V_g=0$)处, S_{yx} (即 E_y)的符号为正,与超导体中的涡旋能斯特信号相同(参考文献15和16)。

$$S_{peak} \sim \frac{k_B \ln 2}{e} \left(\frac{1}{n + \frac{1}{2}} \right) \quad (4)$$

在低 T 下, S 与 H 和 T 无关。

在图2中, S 在 $n=-1$ LL处的峰值从 5 T时的 25 增加到 9 和 14 T时的 $41 \mu\text{V/K}$ 。此外,随着 T 从 20 K增加到 50 K(图c),峰值仅微弱增加($41-48 \mu\text{V}$),这与 $H=0$ 时 T 的线性行为形成鲜明对比(图1,插图)。这与GJ的预测一致,即 S (在峰值处)在足够低的 T 下会饱和至一个与 T 和 H 无关的值。这种饱和现象与 S 在 $H=0$ 时 T 的线性行为形成对比[图1(c)]。

然而在石墨烯中,贝里相位效应会导致 $\frac{1}{2}$ -LL索引中的整数位移。³在评估 $\sigma_{xy} \sim \sum_n f(E_n)$ 时, $\frac{1}{2}$ -整数位移意味着 $S_{peak}(n)$ 随着 $k_B \ln 2 / (en)$ 而减小,而非等式(4)。在 9 T下,当 $n=-1$ 时,测量值 $S_{peak}=41 \mu\text{V/K}$ 已经略微超过了等式(4)中预测的值 $39.7 \mu\text{V/K}$ 。对更纯净样品的改进测量可能会得到更接近狄拉克系统预测值 $59.6 \mu\text{V/K}$ 的数值。原则上,等式(4)提供了一种在微米尺度上以极高分辨率测量 δT 的方法——其分辨率可与电阻测量相媲美。

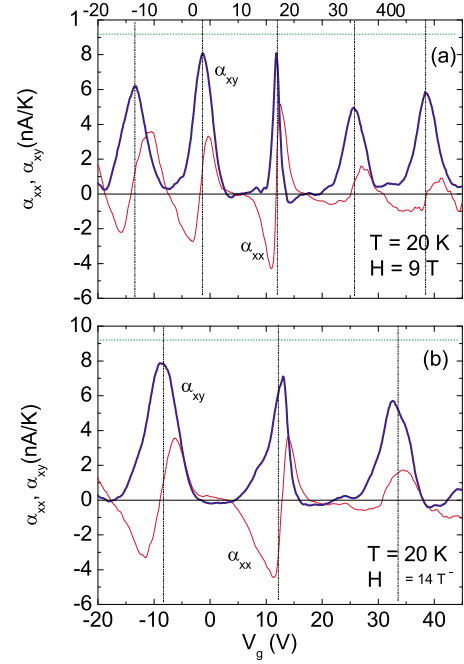


图4. (彩色在线) 热电响应函数 α_{xx} (淡色曲线)和 α_{xy} (粗体)随 V_g 的变化曲线,分别在 9 T(a)和 14 T(b)条件下, $T=20$ K,由 S 和 S_{yx} 计算得出[等式(2)]。在(a)中, $n=0$ 处的峰宽比其他峰窄约4倍。两幅图中,垂直线标定了 G_{xx} 的峰值位置。水平虚线为 $(4k_B e/h) \ln 2$ 。

Nernst信号 S_{yx} 的曲线在图3中展示了 $H=5$ 、 9 和 14 T时的情况。在三个面板中,主要特征是位于 $n=0$ LL处的大型中心峰,该峰高于 $n=0$ 能级的峰。(相比之下,图2中的 S 在 $n=0$ 处被抑制,但在 $n=0$ 时较大。)我们注意到,观测到的大型Nernst峰以及 S 在 $n=0$ 处的强烈抑制与针对二次色散系统的平顶轮廓计算结果存在显著不一致。⁹ S_{yx} 在 $n=0$ 处的正号意味着Nernst E 场 $\mathbf{E}_N$ 与 $\mathbf{H} \times (-\nabla T)$ 平行(与超导体涡旋-Nernst效应中的符号相同)。^{15,16}

在测量了 S 和 S_{yx} 后,我们可以使用等式(2)来计算热电导率张量元素 α_{ij} 。我们在图4(a)(对于 $H=9$ T)和图4(b)(14 T)中展示了 α_{xy} (粗曲线)和 α_{xx} (细曲线)随 V_g 的变化关系。与 S 和 S_{yx} 不同, α_{xy} 中的峰值要窄得多,并且在 α_{xy} 名义上为零的宽区间内明显分离。

α_{xy} 的一个显著特征是其峰值名义上与 n 无关。GJ模型(参考文献8和9)的预测是:[在图1(a)的几何结构中],当 μ 与体内的 E_n 对齐时(对于 $n=0$),横向电流 J_y 会呈现一系列窄脉冲。单位温度梯度的峰值电流仅取决于普适常数 $k_B e/h$ 的组合。图4验证了 $H=9$ 和 14 T时的这一行为。脉冲的平均值约为 75 nA/K ,与 $(gk_B e/h) \ln 2$ 的20%以内(其中 $g=4$,水平虚线)。考虑到电压引线间梯度 $-\nabla T$ 的估计存在较大不确定性,该一致性令人满意。

然而, α_{xy} 在 $n=0$ 处的峰值引发了一些有趣的问题

questions. As shown in Fig. 1(a), it is not clear how the edge-current calculation of GJ (Refs. 8 and 9) is to be generalized to the $n=0$ LL, which is neither holelike nor electronlike. Hence there is actually no prediction of what the peak value of α_{xy} at $n=0$ should be. Our observation that it also equals $\sim(gk_B/h)\ln 2$ is in need of theoretical explanation.

A bigger puzzle appears when we consider the physical meaning of the area $\mathcal{A}_n(T)$ under each α_{xy} peak. As noted in Eq. (3), the computed profile of α_{xy} vs μ is a peak whose width increases linearly with T . The integral with respect to μ is readily performed to give $\int d\mu \alpha_{xy} = (c_0 g k_B^2 e T)/h$, independent of n ($c_0 \sim 3.29$ is a constant). Noting that α_{xy} in Fig. 4 is plotted vs $V_g \sim n_{2D}$ (the 2D carrier density) rather than μ , we have

$$\mathcal{A}_n(T) = \int dn_{2D} \alpha_{xy} \sim \mathcal{N}_n \frac{c_0 g k_B^2 e T}{h}, \quad (n \neq 0), \quad (5)$$

where $\mathcal{N}_n$ is the peak value of the density of states of the LL of index n . At fixed T , the area under each peak in Fig. 4 is proportional to $\mathcal{N}_n$.

In Fig. 4, it is quite apparent that the width of α_{xy} at $n=0$ is anomalously narrow by a factor of ~ 4 compared with its neighbors. As the peaks attain the same height, the area at $n=0$ is smaller by the same ratio. If we extend Eq. (5) to $n=0$, this implies that $\mathcal{N}_n$ is much smaller for $n=0$ than for $n \neq 0$, which seems unlikely given that states are conserved. The results for α_{xy} uncover a puzzle that resides in the $n=0$ LL— α_{xy} attains a peak value equal to that of its neighbors at $n \neq 0$, but the area enclosed is anomalously small.

These problems underscore further our poor understanding of the $n=0$ LL at the Dirac point. Are the currents carried by edge states for $n=0$? Are there counterpropagating edge states? The thermoelectric response issues are related to problems revealed by resistance experiments. We expect them to be deeply pertinent to the current debate on the nature of the high-field ground state at the Dirac point.^{6,7,10–13,17,18}

We are grateful to P. A. Lee for many discussions. The work was supported by NSF through a MRSEC grant (Grant No. DMR-0819860).

- ¹K. S. Novoselov, D. Jiang, F. Schedin, T. J. Booth, V. V. Khotkevich, S. V. Morozov, and A. K. Geim, Proc. Natl. Acad. Sci. U.S.A. **102**, 10451 (2005).
- ²K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, M. I. Katsnelson, I. V. Grigorieva, S. V. Dubonos, and A. A. Firsov, Nature (London) **438**, 197 (2005).
- ³Y. Zhang, J. Tan, H. L. Stormer, and P. Kim, Nature (London) **438**, 201 (2005).
- ⁴Y. Zhang, Z. Jiang, J. P. Small, M. S. Purewal, Y.-W. Tan, M. Fazlollahi, J. D. Chudow, J. A. Jaszczak, H. L. Stormer, and P. Kim, Phys. Rev. Lett. **96**, 136806 (2006).
- ⁵Z. Jiang, Y. Zhang, H. L. Stormer, and P. Kim, Phys. Rev. Lett. **99**, 106802 (2007).
- ⁶J. G. Checkelsky, L. Li, and N. P. Ong, Phys. Rev. Lett. **100**, 206801 (2008).
- ⁷J. G. Checkelsky, L. Li, and N. P. Ong, Phys. Rev. B **79**, 115434 (2009).

- ⁸S. M. Girvin and M. Jonson, J. Phys. C **15**, L1147 (1982).
- ⁹M. Jonson and S. M. Girvin, Phys. Rev. B **29**, 1939 (1984).
- ¹⁰H. A. Fertig and L. Brey, Phys. Rev. Lett. **97**, 116805 (2006).
- ¹¹D. A. Abanin, P. A. Lee, and L. S. Levitov, Phys. Rev. Lett. **96**, 176803 (2006).
- ¹²D. A. Abanin, K. S. Novoselov, U. Zeitler, P. A. Lee, A. K. Geim, and L. S. Levitov, Phys. Rev. Lett. **98**, 196806 (2007).
- ¹³E. Shimshoni, H. A. Fertig, and G. V. Pai, Phys. Rev. Lett. **102**, 206408 (2009).
- ¹⁴J. P. Small, K. M. Perez, and P. Kim, Phys. Rev. Lett. **91**, 256801 (2003).
- ¹⁵Y. Wang, L. Li, and N. P. Ong, Phys. Rev. B **73**, 024510 (2006).
- ¹⁶In our view, the sign reported in Refs. 17 and 18 is incorrect.
- ¹⁷Y. M. Zuev, W. Chang, and P. Kim, Phys. Rev. Lett. **102**, 096807 (2009).
- ¹⁸P. Wei, W. Bao, Y. Pu, C. N. Lau, and J. Shi, Phys. Rev. Lett. **102**, 166808 (2009).

问题。如图1(a)所示, 尚不清楚如何将GJ的边缘电流计算(参考文献8和9)推广到既非空穴型也非电子型的 $n=0$ LL。因此实际上无法预测 $n=0$ 时 α_{xy} 的峰值应为多少。我们观察到其还等于 $\sim (gk_B/h) \ln 2$, 这一现象尚需理论解释。

当我们考虑每个 α_{xy} 峰值下面积 $\mathcal{A}_n(T)$ 的物理意义时, 会出现一个更大的谜题。如等式(3)中所述, α_{xy} 与 μ 的计算轮廓是一个宽度随 T 线性增加的峰值。关于 μ 可轻松求得 $\int d\mu \alpha_{xy} = (c_0 g k_B^2 e T) / h$, 且与 n 无关($c_0 \sim 3.29$ 为常数)。注意到图4中的 α_{xy} 是绘制于 $V_g \sim n_{2D}$ (二维载流子密度)而非 μ 上, 我们有

$$\mathcal{A}_n(T) = \int dn_{2D} \alpha_{xy} \sim \mathcal{N}_n \frac{c_0 g k_B^2 e T}{h}, (n=0), (5)$$

其中 $\mathcal{N}_n$ 是指数 n 的朗道能级态密度的峰值。在固定 T 时, 图4中每个峰下的面积与 $\mathcal{N}_n$ 成正比。

在图4中, 很明显 α_{xy} 在 $n=0$ 处的宽度比其相邻区域异常地窄约4倍。当峰达到相同高度时, $n=0$ 处的面积也以相同比例缩小。如果我们把等式(5)延伸到 $n=0$, 这意味着 $\mathcal{N}_n$ 在 $n=0$ 处比 $n=0$ 处小得多, 这在状态守恒的情况下似乎不太可能。 α_{xy} 的结果揭示了一个谜题: 在 $n=0$ 处, LL- α_{xy} 达到的峰值与其相邻区域在 $n=0$ 处的峰值相等, 但所包围的面积异常地小。

这些问题进一步凸显了我们对狄拉克点处 $n=0$ 朗道能级的了解不足。边缘态是否承载 $n=0$ 的电流? 是否存在反向传播的边缘态? 热电响应问题与电阻实验揭示的问题相关。我们预计这些问题将深刻影响当前关于狄拉克点高场基态本质的争论。^{6,7,10-13,17,18}

我们感谢P. A. Lee的多次讨论。本研究由美国国家自然科学基金会(NSF)通过MRSEC资助项目(资助编号: DMR-0819860)支持。

¹ K. S. Novoselov, D. Jiang, F. Schedin, T. J. Booth, V. V. Khotkevich, S. V. Morozov 和 A. K. Geim, 《美国国家科学院院刊》**102**, 10451 (2005年)。

² K. S. 诺沃肖洛夫, A. K. 盖姆, S. V. 莫罗佐夫, D. 江, M. I. 卡茨内尔森, I. V. 格里戈里耶娃, S. V. 杜博诺斯和A. A. 菲尔索夫, 《自然》(伦敦)**438**, 197 (2005年)。

³ Y. Zhang, J. Tan, H. L. Stormer 和 P. Kim, 《自然》(伦敦)**438**, 201 (2005年)。

⁴ 张宇、蒋志、J. P. Small, M. S. Purewal, 谭永伟, M. Fazlollahi, J. D. Chudow, J. A. Jaszczak, H. L. Stormer 和 P. Kim, 《物理评论快报》**96**卷, 136806页 (2006年)。

⁵ Z. Jiang, Y. Zhang, H. L. Stormer 和 P. Kim, 《物理评论快报》**99**, 106802 (2007年)。

⁶ J. G. Checkelsky, L. Li 和 N. P. Ong, 《物理评论快报》**100**卷, 206801页 (2008年)。

⁷ J. G. Checkelsky, L. Li 和 N. P. Ong, 《物理评论B》**79**, 115434 (2009年)。

⁸ S. M. 吉尔文与 M. 约翰逊, 《物理化学杂志》**15**卷, L1147页 (1982年)。

⁹ M. Jonson 和 S. M. Girvin, 《物理评论B辑》**29**, 1939年 (1984年)。

¹⁰ H. A. Fertig 与 L. Brey, 《物理评论快报》**97**卷, 116805页 (2006年)。

¹¹ D. A. 阿巴宁、P. A. 李和 L. S. 利维托夫, 《物理评论快报》**96**卷, 176803页 (2006年)。

¹² D. A. 阿巴宁、K. S. 诺沃肖洛夫、U. 茨特勒、P. A. 李、A. K. 盖姆和 L. S. 利维托夫, 《物理评论快报》**98**, 196806 (2007年)。

¹³ E. Shimshoni, H. A. Fertig 和 G. V. Pai, 《物理评论快报》**102**卷, 206408页 (2009年)。

¹⁴ J. P. Small, K. M. Perez 和 P. Kim, 《物理评论快报》**91**卷, 256801页 (2003年)。

¹⁵ Y. Wang, L. Li 和 N. P. Ong, 《物理评论B》**73**卷, 024510页 (2006年)。

¹⁶ 我们认为, 文献17和18中报道的体征是错误的。

¹⁷ Y. M. Zuev, W. Chang 和 P. Kim, 《物理评论快报》**102**卷, 096807页 (2009年)。

¹⁸ P. Wei, W. Bao, Y. Pu, C. N. Lau 和 J. Shi, 《物理评论快报》**102**卷, 166808页 (2009年)。